

Recommender Systems

Individual Project

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Music or Movies?

Let's build a recommender prototype!

A progressive, week-by-week implementation project.

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Purpose

Build one recommender prototype over the course, expanding it every week as new techniques are introduced.

- **Learn by building:** Every method becomes a working module.
- **Compare methods:** Same data, multiple algorithms.
- **Think critically:** Accuracy is not enough: analyse diversity, novelty, bias, scalability and other aspects you may think they are relevant.

Choose your track: Music or Movies

Movie track

Main Datasets: Movielens

Other: Movielens 100k

Music track

Main Datasets: Last.fm HetRec 2011

Other: MSD Taste Profile, Spotify MPD

You can explore alternative data sources. Cite the dataset source, read the license/README, and document any filtering or sampling.

Week by Week development

Suggested agenda:

- Design implement prototype user interface
- Setup + dataset + EDA + preprocessing
- Non personalised
- Collaborative based filtering
- Content based filtering
- Matrix factorisation
- Evaluation

Use Agile development (basic evaluation from the beginning)

Deliverables and Assessment

Deliverables:

- **Prototype**
- **Slide deck** (technical challenges, method comparison, final remarks)

Evaluation

- Technical method implementations — 50%
- Evaluation — 30%
- User Experience — 20%

Ressources

MOVIES

[GroupLens MovieLens datasets](#)

[MovieLens Latest Small](#)

[Datasets MovieLens catalog](#)

MUSIC

[GroupLens HetRec 2011 / last.fm](#)

[Song Dataset Taste Profile subset](#)

[Spotify Research Million Playlist Dataset](#)

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